Their defining greatest-lower-bound and least-upper-bound formulas make these functions definable over the original parameter base. Thus

$$t \longmapsto \gamma_t = \max\{|\alpha_t|, |\beta_t|, 0\}$$

is a definable Γ-valued function on A_{RV}. By Lemma 2.41, its image is doubly bounded. A common upper bound for the γ_t bounds $p(A)$ in both directions, proving the lemma. □

Resolved referee R044 [Gap] – Bounds for an open-disc partition.
Original source line 1152. The lower bound has a short direct proof: choose η with $A \subseteq \mathcal{O}_\eta^n$. If $p(a) < \eta$ for some a, then $A \subseteq \mathfrak{o}(a, p(a))$, forcing p to be constant; otherwise $p(A) \subseteq [\eta, \infty)$. For the upper-bound argument, justify the nonempty finite choices in $A_{\mathrm{val}(c)}$ using the corrected curve-selection/model argument, and use the nonempty finite limit supplied by the limit lemma. In the general case, take a definable bound for each fiber image $p(A_t)$ by its definable endpoints in Γ, then apply Lemma 2.41; fiberwise boundedness by itself is not the uniform conclusion.

Author resolution. The author agrees that the original proof is quick and the comments spell out details. The expansion retains its lower-bound, limit-point contradiction and uniform-fiber-bound strategy. It gives the direct lower-bound calculation, justifies finite nonempty choices over a temporary base, and writes out the definable endpoint bound used with Lemma 2.41. The statement is unchanged.

This lemma, in its various incarnations, is crucial for the good behavior of motivic Fourier transform (see [8, § 11] and [20]). For essentially the same reason, the main construction of this paper depends heavily on it (often in combination with Lemma 2.41).

Lemma 2.54. *Let $A \subseteq \mathrm{VF}^n$ be a definable set. Suppose that, for each $a \in A_1$, the fiber $A_a \subseteq \mathrm{VF}^{n-1}$ is open (respectively, closed). Then there is a finite definable set $A' \subseteq A_1$ such that $A \cap ((A_1 \smallsetminus A') \times \mathrm{VF}^{n-1})$ is open (respectively, closed) in $(A_1 \smallsetminus A') \times \mathrm{VF}^{n-1}$.*

Proof. We prove the open case over an arbitrary definable unary base $I \subseteq \mathrm{VF}$, allowing empty fibers, with $A \subseteq I \times \mathrm{VF}^{n-1}$. The closed case then follows by taking the relative complement in $I \times \mathrm{VF}^{n-1}$. In the statement take $I = A_1$.

Put $X = I \times \mathrm{VF}^{n-1}$ and let

$$S = A \cap \mathrm{Cl}_X(X \setminus A).$$

Thus no point of S is interior to A relative to X. It is enough to show that S_1 is finite. For each $\gamma \in \Gamma$, set

$$T_\gamma = \{(a, x) \in S : \mathfrak{o}(x, \gamma) \subseteq A_a\}.$$

Openness of every fiber gives

$$S_1 = \bigcup_{\gamma \in \Gamma} (T_\gamma)_1.$$

If every $(T_\gamma)_1$ were finite, this definable Γ-indexed union would be finite as well: lift the index to RV and use the RV-indexed VF-dimension bound [17, Lemma 4.3]. Consequently, if S_1 is infinite, some $(T_\gamma)_1$ contains an open disc D.

For this geometric argument we may temporarily enlarge the parameter base to a small VF-generated base containing the data, as in Lemma 2.48. For every $a \in D$, the nonempty definable fiber $(T_\gamma)_a$ has an algebraic point by elementarity of the algebraic-closure model over this base together with a. Compactness therefore gives a definable finitary function

$$F : D \longrightarrow \mathcal{P}(\mathrm{VF}^{n-1}), \qquad \varnothing \neq F(a) \subseteq (T_\gamma)_a.$$

By Lemma 2.44, choose $a_0 \in D$ at which F is continuous, and put $E = F(a_0)$. After shrinking D around a_0, continuity gives

$$F(a) \subseteq \bigcup_{e \in E} \mathfrak{o}(e, \gamma) \qquad (a \in D).$$

Here E is finite and nonempty. The finitely many definable sets

$$D_e = \{a \in D : F(a) \cap \mathfrak{o}(e, \gamma) \neq \varnothing\}, \qquad e \in E,$$

cover D. Hence one of them contains an open subdisc D'. Fix its index e. For each $a \in D'$, choose $x \in F(a) \cap \mathfrak{o}(e, \gamma)$. Since $(a, x) \in T_\gamma$ and intersecting open polydiscs of the same radius are equal,

$$\mathfrak{o}(e, \gamma) = \mathfrak{o}(x, \gamma) \subseteq A_a.$$

Thus $D' \times \mathfrak{o}(e, \gamma)$ is an open subset of X contained in A and meeting S, a contradiction. Therefore S_1 is finite. Away from S_1, every point of A is interior relative to X, which proves the open case and, by relative complementation, the closed case. □

Resolved referee R045 [Gap; revised] – The fiberwise openness proof needs reconstruction. *Original source line 1173.* The displayed proof contains genuine mistakes, but the conclusion has a repair. Retain $d = \dim_{\mathrm{VF}}(S) < n$ instead of replacing it by $n - 1$. Define

$$\alpha(a, x) = \min\{\beta \geq 0 : \mathfrak{o}(x, \beta) \subseteq A_a\} \quad ((a, x) \in S).$$

This minimum exists: for a nonempty complement it is the maximum of 0 and the supremum of the valuative distances from x to that complement; for a full fiber it is 0. The original unrestricted minimum fails for full fibers. If S_1 is infinite, choose a smooth algebraic chart of a stratum of S whose projection to the first coordinate is nonconstant. In characteristic zero, a separating coordinate projection, including the first coordinate, is generically étale; a local branch over an open polydisc is therefore definable and continuous. Pull α back to this chart and shrink to an open polydisc where it is constant, say ϵ, using generic local constancy of maps into Γ. Fix the remaining chart coordinates and shrink the first-coordinate disc D until the resulting section $a \mapsto (a, s(a)) \in S$ satisfies $s(a) \in \mathfrak{o}(s(a_0), \epsilon)$. Then

$$D \times \mathfrak{o}(s(a_0), \epsilon) \subseteq A,$$

contradicting $(a_0, s(a_0)) \in S$. This proves that S_1 is finite, including when $d < n - 1$. The contradiction concerns interior points of A, not an open polydisc inside S. For the closed case and the next proposition, formulate the lemma over a fixed definable base $I \subseteq \mathrm{VF}$, allowing empty fibers; passage between I and the actual projection costs only its finite unary boundary.

Author resolution. The author approves the independently reconstructed bad-locus proof. A definable family of valid radii and the RV-indexed union principle supply one radius on a whole disc of base points. Finite nonempty choices and unary generic continuity give an open subset of A meeting S, the required contradiction. No codimension-one assumption or minimum-radius choice is needed. The statement is retained, and working over a fixed unary base with empty fibers allowed justifies the closed case by relative complement.

Proposition 2.55. *Let A be as in Lemma 2.54 and $f : A \longrightarrow \mathrm{VF}$ a definable function such that for each $a \in A_1$, the induced function $f_a : A_a \longrightarrow \mathrm{VF}$ is continuous. Then there is a finite definable set $A' \subseteq A_1$ such that f is continuous on $\bigcup_{a \in A_1 \smallsetminus A'} a \times A_a$.*

For the closed-graph argument, the projective line $\mathbb{P}^1(\mathrm{VF})$ is understood with the valuation topology, as in Lemma 2.50.

Proof. Put $I = A_1$ and write H_a for the graph of f_a. Let

$$G_a = \mathrm{Cl}_{\mathrm{VF}^{n-1}\times\mathbb{P}^1(\mathrm{VF})}(H_a), \qquad G = \bigcup_{a\in I}\{a\}\times G_a.$$

This family is definable: fiberwise closure is expressed uniformly by the valuative neighborhood formula in each affine chart. Every G_a is closed in $\mathrm{VF}^{n-1}\times\mathbb{P}^1(\mathrm{VF})$.

Apply the fixed-base version proved in Lemma 2.54, with empty fibers allowed, to G in each of the two standard affine charts of $\mathbb{P}^1(\mathrm{VF})$. Outside the union I' of the resulting two finite definable exceptional sets, G is closed in

$$(I\setminus I')\times\mathrm{VF}^{n-1}\times\mathbb{P}^1(\mathrm{VF}),$$

since closedness is local on this open chart cover. Fiberwise continuity gives

$$H_a = G_a\cap\big(A_a\times\mathbb{P}^1(\mathrm{VF})\big).$$

Consequently, the graph of the restricted function is closed relative to

$$\big(A\cap((I\setminus I')\times\mathrm{VF}^{n-1})\big)\times\mathbb{P}^1(\mathrm{VF}).$$

The closed-graph criterion (Lemma 2.50) proves the asserted continuity. This spells out the relative-closure step in the original implication. □

Resolved referee R046 [Gap; revised] – Relative graph closedness is insufficient for the cited lemma.

Original source line 1188. Continuity of f_a makes its graph closed in $A_a\times\mathbb{P}^1(\mathrm{VF})$, whereas the preceding closed-fiber lemma concerns ambient closed fibers. Define instead

$$G_a = \overline{\mathrm{graph}(f_a)}\subseteq\mathrm{VF}^{n-1}\times\mathbb{P}^1(\mathrm{VF}).$$

Use the preceding lemma in its fixed-base form over $I = A_1$, allowing empty fibers, in each of the two affine charts of $\mathbb{P}^1$. Equivalently, also discard the finite unary boundaries of the chart projections when applying the lemma as currently stated. Outside the union of these finite exceptional sets, G is relatively closed over I. Fiberwise continuity gives $\mathrm{graph}(f) = G\cap(A\times\mathbb{P}^1)$, so the graph is closed in $A\times\mathbb{P}^1$, and the compactified closed-graph criterion applies. This proves the proposition without an open/closed-fiber hypothesis on A_a; clarify the phrase 'as in Lemma 2.54' accordingly.

Author resolution. The author approves expanding the original implication through its relative-closure step. Ambient closures of the fiber graphs form a definable closed-fiber family; Lemma 2.54 applies in the two affine charts over the fixed unary base. Intersecting back with the domain gives relative graph closedness, so Lemma 2.50 gives continuity. The statement and its existing scope are retained. The target wording now explicitly uses the projective line with the valuation topology.

This is analogous to [4, Corollary 6.2.4] in the o-minimal environment and [22, Lemma 2.38] in the T-convex environment.

3. Hrushovski-Kazhdan style integral

Definition 3.1. The Grothendieck semigroup $\mathbf{K}_+\mathcal{C}$ of a category $\mathcal{C}$ endowed with a binary operation "$\smallsetminus$" and a binary relation "$\subseteq$" (subobject) is the free semigroup generated by the isomorphism classes of $\mathcal{C}$, subject to the usual scissor relation $[A\smallsetminus B]+[B]=[A]$, when $B\subseteq A$, where $[A]$, $[B]$ denote the isomorphism classes of the objects A, B, and "$\smallsetminus$" and "$\subseteq$" are compatible with isomorphisms between objects, usually they are just set subtraction and set inclusion. Sometimes $\mathcal{C}$ is also equipped with a binary operation (for example, cartesian product) that induces multiplication in $\mathbf{K}_+\mathcal{C}$, in which case $\mathbf{K}_+\mathcal{C}$ becomes a (commutative) semiring. The formal groupification of $\mathbf{K}_+\mathcal{C}$, which is then a ring, is denoted by $\mathbf{K}\mathcal{C}$.